

LU - Decomposition (Factorization)

Def 11.1 Lower and upper triangular matrices are matrices L and U respectively such that $L_{i,j} = 0$ for $i > j$ and $U_{i,j} = 0$ for $i < j$.

$$U = \begin{bmatrix} * & 0 & 0 & 0 \\ 0 & * & 0 & 0 \\ 0 & 0 & * & 0 \\ 0 & 0 & 0 & * \end{bmatrix} \quad L = \begin{bmatrix} * & 0 & 0 & 0 \\ * & * & 0 & 0 \\ * & * & * & 0 \\ * & * & * & * \end{bmatrix}$$

Def 11.2 The diagonal of A denoted by $\text{diag}(A)$ is the set

$$\text{diag}(A) = \{A_{1,1}, A_{2,2}, \dots, A_{n,n}\}$$

Note that $A_{i,i} = \langle \vec{e}_i, A\vec{e}_i \rangle$

Theorem 11.3 (Uniqueness of LU-factorization)

Let $A \in \mathbb{R}^{n \times n}$ be invertible and let $A = LU$ where L and U are lower and upper triangular matrices such that $\text{diag}(L) = \{1, \dots, 1\}$.

Then the above factorization is unique and is known as the LU-factorization (decomp.) of A .

Computing LU

By Gaussian elimination:

$$\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \xrightarrow{\substack{+\lambda_1 r_1 \\ +\lambda_2 r_1}} \begin{bmatrix} * & * & * \\ 0 & * & * \\ 0 & * & * \end{bmatrix} \xrightarrow{+\lambda_3 r_2} \begin{bmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \end{bmatrix} = U$$

Def 11.4 An elementary matrix, denoted as

$E_{j,i}(\lambda_{j,i})$ is of the form

$$E_{j,i}(\lambda_{j,i}) = I + \begin{matrix} & & i \\ & & | \\ j & \begin{bmatrix} -\lambda_{j,i} & \\ & \end{bmatrix} \\ & & i \end{matrix}$$

where $j > i$.

Properties 11.5

letting $A = \begin{bmatrix} - & A_{1,\cdot} & - \\ & \vdots & \\ - & A_{n,\cdot} & - \end{bmatrix}$

$$i. \quad E_{j,i}(\lambda_{j,i}) A = \begin{bmatrix} A_{1,\cdot} \\ \vdots \\ A_{j,\cdot} + \lambda_{j,i} A_{i,\cdot} \\ \vdots \\ A_{n,\cdot} \end{bmatrix}$$

$$r_j' = \lambda_{j,i} r_i + r_j$$

$$ii. \quad E_{j,i}^{-1}(\lambda_{j,i}) = E_{j,i}(-\lambda_{j,i})$$

Examples :

$$i. \quad \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 3 & 1 \\ 3 & 4 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & -1 & 1 \\ 3 & 4 & 5 \end{bmatrix}$$

$$ii. \quad \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$iii. \quad \prod_{i=1}^{n-1} \prod_{j>i} E_{j,i}^{-1}(\lambda_{j,i}) = E_{2,1}^{-1}(\lambda_{2,1}) E_{3,1}^{-1}(\lambda_{3,1}) \dots E_{n,1}^{-1}(\lambda_{n,1}) E_{3,2}^{-1}(\lambda_{3,2}) \dots$$

$$= \begin{bmatrix} 1 & 0 & 0 & \dots & 0 & 0 \\ -\lambda_{2,1} & 1 & 0 & \dots & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} -\lambda_{3,1} & -\lambda_{3,2} & 1 & \dots & 0 & 0 \\ & \vdots & \ddots & \ddots & \ddots & \vdots \\ -\lambda_{n,1} & -\lambda_{n,2} & -\lambda_{n,3} & \dots & -\lambda_{n,n-1} & 1 \end{bmatrix}$$

For example given 3×3 elementary matrices

$E_{2,1}(\lambda_{2,1})$, $E_{3,1}(\lambda_{3,1})$, and $E_{3,2}(\lambda_{3,2})$

such that $(E_{3,2} E_{3,1} E_{2,1})A = U$

then $A = LU$ where $L = \begin{bmatrix} 1 & 0 & 0 \\ -\lambda_{2,1} & 1 & 0 \\ -\lambda_{3,1} & -\lambda_{3,2} & 1 \end{bmatrix}$

LU - factorization by Algebra : set $A = LU$
where $\text{diag}(L) = \{1, \dots, 1\}$

$$\begin{bmatrix} A_{1,1} & A_{1,2} & A_{1,3} \\ A_{2,1} & A_{2,2} & A_{2,3} \\ A_{3,1} & A_{3,2} & A_{3,3} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ L_{2,1} & 1 & 0 \\ L_{3,1} & L_{3,2} & 1 \end{bmatrix} \begin{bmatrix} U_{1,1} & U_{1,2} & U_{1,3} \\ 0 & U_{2,2} & U_{2,3} \\ 0 & 0 & U_{3,3} \end{bmatrix}$$

Theorem 11.6 Assume that A is invertible
so that $A\vec{x} = \vec{b}$ has unique solution $\vec{x}_0$.
Then $\vec{x}_0$ satisfies the two equations

$$i. \quad U \vec{x} = \vec{c}$$

$$ii. \quad L \vec{c} = \vec{b}$$

Exercise:

1) Find the LU decomposition of

$$A = \begin{bmatrix} 1 & 3 & 7 \\ 2 & 1 & -4 \\ -3 & 1 & -2 \end{bmatrix}$$

and use it to solve $A\vec{x} = \vec{b}$ where

$$\vec{b} = \begin{bmatrix} 14 \\ 0 \\ -3 \end{bmatrix}$$

$$\begin{bmatrix} \textcircled{1} & 3 & 7 \\ 2 & 1 & -4 \\ -3 & 1 & -2 \end{bmatrix} \begin{matrix} -2r_1 \\ +3r_1 \end{matrix} \rightarrow \begin{bmatrix} 1 & 3 & 7 \\ 0 & \textcircled{-5} & -18 \\ 0 & 10 & 19 \end{bmatrix} \begin{matrix} \\ +2r_2 \end{matrix}$$

$$\rightarrow \begin{bmatrix} 1 & 3 & 7 \\ 0 & -5 & -18 \\ 0 & 0 & -17 \end{bmatrix} = U$$

$\therefore A = LU$ where U is above and

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & -2 & 1 \end{bmatrix}$$

Now solving $A\vec{x} = \vec{b}$:

$$L\vec{c} = \vec{b} : \left[\begin{array}{ccc|c} 1 & 0 & 0 & 14 \\ 2 & 1 & 0 & 0 \\ -3 & -2 & 1 & -3 \end{array} \right]$$

$$c_1 = 14$$

$$c_2 = -2c_1 = -28$$

$$c_3 = -3 + 3c_1 + 2c_2 = -17$$

$$U\vec{x} = \vec{c} : \left[\begin{array}{ccc|c} 1 & 3 & 7 & 14 \\ 0 & -5 & -18 & -28 \\ 0 & 0 & -17 & -17 \end{array} \right]$$

$$x_3 = 1$$

$$x_2 = 2$$

$$x_1 = 1$$

$$\therefore \vec{x} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

LU with Partial Pivoting

If A cannot be transformed to U under products of elementary matrices, a permutation

matrix may be applied to interchange rows

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} * & * & * \\ 0 & 0 & * \\ 0 & * & * \end{bmatrix} \xrightarrow{\text{blue arrow}} \begin{bmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \end{bmatrix}$$

$$\boxed{PA = LU}$$

LDU Decomposition

Given upper diagonal matrix U we can write $U = DU'$ where D is a diagonal and U' is also upper triangular with ones across its diagonal.

For example given $U = \begin{bmatrix} u_{1,1} & u_{1,2} & u_{1,3} \\ 0 & u_{2,2} & u_{2,3} \\ 0 & 0 & u_{3,3} \end{bmatrix}$

we have that

$$U = \begin{bmatrix} u_{1,1} & & \\ & u_{2,2} & \\ & & u_{3,3} \end{bmatrix} \begin{bmatrix} 1 & \frac{u_{1,2}}{u_{1,1}} & \frac{u_{1,3}}{u_{1,1}} \\ 0 & 1 & \frac{u_{2,3}}{u_{2,2}} \\ 0 & 0 & 1 \end{bmatrix}$$

Exercise :

2) Find the LDU decomposition of the matrix

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 3 & 1 \\ -4 & 6 & -2 \end{bmatrix}$$

$$\begin{bmatrix} \textcircled{1} & 0 & 3 \\ 2 & 3 & 1 \\ -4 & 6 & -2 \end{bmatrix} \xrightarrow[-4r_1]{-2r_1} \begin{bmatrix} 1 & 0 & 3 \\ 0 & \textcircled{3} & -5 \\ 0 & 6 & 10 \end{bmatrix} \xrightarrow{-2r_2}$$

$$\begin{bmatrix} 1 & 0 & 3 \\ 0 & 3 & -5 \\ 0 & 0 & 20 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & & \\ & 3 & \\ & & 20 \end{bmatrix}}_D \underbrace{\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -5/3 \\ 0 & 0 & 1 \end{bmatrix}}_U$$

$$\therefore A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -4 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ & 3 & \\ & & 20 \end{bmatrix} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -5/3 \\ 0 & 0 & 1 \end{bmatrix}$$

Theorem 11.7 Let A be symmetric and invertible and let $A = LDU$ be the (unique) LDU - factorization. Then $U = L^T$ or $L = U^T$.

Proof: Since $A^T = A$, $LDU = (LDU)^T = U^T D L^T$
 and thus $L = U^T$ and $U = L^T$ by the
 uniqueness of LDU

Exercise:

3) Find the LDU decomposition of the
 symmetric matrix

$$A = \begin{bmatrix} 2 & 1 & -5 \\ 1 & 3 & -2 \\ -5 & -2 & 13 \end{bmatrix}$$

$$\begin{bmatrix} \textcircled{2} & 1 & -5 \\ 1 & 3 & -2 \\ -5 & -2 & 13 \end{bmatrix} \xrightarrow[\substack{-\frac{1}{2}r_1 \\ +\frac{5}{2}r_1}]{\substack{-\frac{1}{2}r_1 \\ +\frac{5}{2}r_1}} \begin{bmatrix} 2 & 1 & -5 \\ 0 & \textcircled{5/2} & 1/2 \\ 0 & 1/2 & 1/2 \end{bmatrix} \xrightarrow{-\frac{1}{5}r_2}$$

$$\begin{bmatrix} 2 & 1 & -5 \\ 0 & 5/2 & 1/2 \\ 0 & 0 & 2/5 \end{bmatrix} = \underbrace{\begin{bmatrix} 2 & & \\ & 5/2 & \\ & & 2/5 \end{bmatrix}}_D \underbrace{\begin{bmatrix} 1 & 1/2 & -5/2 \\ 0 & 1 & 1/5 \\ 0 & 0 & 1 \end{bmatrix}}_{U=L^T}$$

$$\therefore A = \begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ -5/2 & 1/5 & 1 \end{bmatrix} \begin{bmatrix} 2 & & \\ & 5/2 & \\ & & 2/5 \end{bmatrix} \begin{bmatrix} 1 & 1/2 & -5/2 \\ 0 & 1 & 1/5 \\ 0 & 0 & 1 \end{bmatrix}$$

Going further we write

$$\begin{bmatrix} 2 & & \\ & 5/2 & \\ & & 2/5 \end{bmatrix} = \begin{bmatrix} \sqrt{2} & & \\ & \sqrt{5/2} & \\ & & \sqrt{2/5} \end{bmatrix}^2$$

and thus

$$\begin{aligned} A &= \begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ -5/2 & 1/5 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{2} & & \\ & \sqrt{5/2} & \\ & & \sqrt{2/5} \end{bmatrix} \left(\begin{bmatrix} 1 & 0 & 0 \\ 1/2 & 1 & 0 \\ -5/2 & 1/5 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{2} & & \\ & \sqrt{5/2} & \\ & & \sqrt{2/5} \end{bmatrix} \right)^T \\ &= \begin{bmatrix} \sqrt{2} & 0 & 0 \\ \sqrt{2}/2 & \sqrt{5/2} & 0 \\ -5\sqrt{2}/2 & \frac{1}{5}\sqrt{5/2} & \sqrt{2/5} \end{bmatrix} \begin{bmatrix} \sqrt{2} & \sqrt{2}/2 & -5\sqrt{2}/2 \\ 0 & \sqrt{5/2} & \frac{1}{5}\sqrt{5/2} \\ 0 & 0 & \sqrt{2/5} \end{bmatrix} \\ &= (L\sqrt{D})(L\sqrt{D})^T \end{aligned}$$

If $A = LDL^T$ and $D_{i,i} > 0$ for all i , then

$A = (L\sqrt{D})(L\sqrt{D})^T$ which is the Cholesky Factorization. This condition is equivalent to A being positive definite.

QR Factorization (See Strang)

Let $A \in \mathbb{R}^{n \times m}$ where $m \leq n$ have linearly independent columns $\{\vec{v}_1, \dots, \vec{v}_m\}$. Using G-S we obtain orthonormal vectors $\{\vec{q}_1, \dots, \vec{q}_m\}$ such that $\vec{v}_k \in \text{span}(\vec{q}_1, \dots, \vec{q}_k)$ for all $1 \leq k \leq m$. Now as

$$\vec{v}_k = \sum_{i=1}^k \langle \vec{v}_k, \vec{q}_i \rangle \vec{q}_i + \sum_{i=k+1}^m 0 \cdot \vec{q}_i$$

we have that

$$\vec{v}_k = [\vec{q}_1, \dots, \vec{q}_k, \dots, \vec{q}_m] \begin{bmatrix} \langle \vec{v}_k, \vec{q}_1 \rangle \\ \langle \vec{v}_k, \vec{q}_2 \rangle \\ \vdots \\ \langle \vec{v}_k, \vec{q}_k \rangle \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Theorem 11.8 (QR-factorization) Let $A \in \mathbb{R}^{n \times m}$ have linearly independent columns $\{\vec{v}_1, \dots, \vec{v}_m\}$ and let $\{\vec{q}_1, \dots, \vec{q}_m\}$ be an orthonormal set obtained by G-S ($\vec{v}_k \in \text{span}(\vec{q}_1, \dots, \vec{q}_k)$)

Then $A = QR$ where

$$Q = [\vec{q}_1 \dots \vec{q}_m]_{n \times m}$$

and $R = \begin{bmatrix} \langle \vec{v}_1, \vec{q}_1 \rangle & \langle \vec{v}_2, \vec{q}_1 \rangle & \dots & \langle \vec{v}_m, \vec{q}_1 \rangle \\ & \langle \vec{v}_2, \vec{q}_2 \rangle & \dots & \langle \vec{v}_m, \vec{q}_2 \rangle \\ & & \ddots & \vdots \\ & & & \langle \vec{v}_m, \vec{q}_m \rangle \end{bmatrix}_{m \times m}$
inverted

This is the reduced QR factorization of A and is unique.